

Monetary Economics and Policy

Pierpaolo Benigno

Chapter 8: New Keynesian
Model with a Banking Sector



Prepared by Severin Rothen

What We Will Learn

- ▶ Why the benchmark New Keynesian model with a **single interest rate** misses key features of monetary transmission once banks and liquidity premia matter.
- ▶ How to introduce a **banking sector** where deposits provide **liquidity services** and government liabilities provide **collateral services**.
- ▶ How these frictions generate a **hierarchy of risk-free rates** (policy rate, deposit rate, credit-market rate) and a wedge between the policy rate and the rate relevant for consumption.
- ▶ How **government liquidity** (reserves and short-term public debt) affects aggregate demand by moving **liquidity premia**, beyond the standard interest-rate channel.
- ▶ How the model delivers a modified **log-linear aggregate demand equation** in which liquidity enters directly and the influence of expected future short rates is **attenuated**.
- ▶ How **optimal liquidity provision** (a Friedman-rule logic) calls for **satiating liquidity** when lump-sum taxes are available, and how this interacts with optimal stabilization policy.

Outline

- 1 Introduction
- 2 Model
- 3 Equilibrium
- 4 Log-linear approximation
- 5 Impulse Responses
- 6 Optimal policy
- 7 Main Takeaways
- 8 Bibliography

- 1 Introduction
- 2 Model
- 3 Equilibrium
- 4 Log-linear approximation
- 5 Impulse Responses
- 6 Optimal policy
- 7 Main Takeaways
- 8 Bibliography

Limitations of the Standard NK Model

- ▶ Benchmark New-Keynesian (NK) model:
 - ▶ Single nominal interest rate i_t for:
 - ▶ Policy rate,
 - ▶ Money-market rate,
 - ▶ Credit-market rate relevant for households' Euler equation.
 - ▶ No explicit banking sector.
- ▶ Great Financial Crisis 2007–2008:
 - ▶ Large spreads between different credit standards, money-market rates, and the policy rate.
 - ▶ Banks crucial for liquidity creation (deposits, reserves) and credit provision to households and firms.

Extended NK Model

- ▶ We embed a banking sector into the NK framework.
- ▶ Two key financial frictions:
 - ▶ **Households:** deposits D_t provide liquidity services, they enter utility via $V(\frac{D_t}{P_t})$.
 - ▶ **Banks:** government liabilities (reserves and treasury bills) provide collateral services for issuing deposits.
- ▶ As a result, deposit rate i_t^D carries a liquidity premium versus credit-market rate i_t .
- ▶ Reserves (and government debt) carry a liquidity premium versus deposit rate i_t^D .

Key Results

- Hierarchy of risk-free rates (in steady state and in equilibrium):

$$i_t^X \leq i_t^D \leq i_t$$

- Transmission of monetary policy:
 - Policy rate i_t^X first affects banks' balance sheets, then affects deposit rate i_t^D and credit-market rate i_t .
 - Households' consumption/saving decisions depend on i_t , not directly on i_t^X .
- Government liquidity:
 - Total government liabilities B_t^g (reserves + treasury bills) affect equilibrium liquidity premia, which changes the wedge between i_t and i_t^X .
 - Hence government liquidity directly influences aggregate demand and inflation.

Policy Implications

- ▶ Central bank has two instruments:
 - ▶ Policy rate on reserves i_t^X ,
 - ▶ Quantity of government liquidity B_t^g (reserves + treasury bills).
- ▶ Government liquidity affects:
 - ▶ Liquidity premia on deposits and reserves,
 - ▶ Credit-market rate i_t ,
 - ▶ Aggregate demand and inflation.
- ▶ Important when central bank balance sheet expands/contracts:
 - ▶ Quantitative easing / balance sheet policies,
 - ▶ Exit from large balance sheets.

- 1 Introduction
- 2 Model**
- 3 Equilibrium
- 4 Log-linear approximation
- 5 Impulse Responses
- 6 Optimal policy
- 7 Main Takeaways
- 8 Bibliography

Banking Sector Balance Sheet

- ▶ At time t , a representative bank has the balance sheet constraint:

$$X_t + B_t^F + B_t = D_t + N_t. \quad (8.1)$$

- ▶ X_t : Central bank reserves, with interest rate i_t^X .
 - ▶ B_t^F : Treasury bills (government debt).
 - ▶ B_t : Short-term private bonds, with interest rate i_t (credit-market rate).
 - ▶ D_t : Deposits, with interest rate i_t^D .
 - ▶ N_t : Bank equity.
- ▶ Treasury debt is fully backed by the central bank, therefore both pay interest rate i_t^X and we define overall government debt as:

$$B_t^g \equiv X_t + B_t^F.$$

Collateral and Zero Lower Bound

- ▶ Intermediaries are subject to collateral requirement:

$$B_t^g \geq \rho D_t \geq 0, \quad 0 \leq \rho \leq 1.$$

- ▶ When $\rho = 1$ we are in the case of narrow banking, i.e., all deposits must be fully backed by reserves.
 - ▶ With $\rho = 0$ there is no collateral requirement.
- ▶ Due to the possibility of converting reserves into cash there is a zero lower bound on reserves:

$$i_t^X \geq 0.$$

Short-term Private Bonds

- ▶ In general B_t can be positive (bank holds bonds) or negative (bank is borrowing).
- ▶ We will show later that $i_t \geq i_t^D$. This is because deposits provide liquidity services and, therefore, receive a liquidity premium whereas private bonds are risk-free but not liquid.

Bank Profits

- ▶ Bank profits at time $t + 1$ are:

$$\Psi_{t+1}^I = (1 + i_t)B_t + (1 + i_t^X)B_t^g - (1 + i_t^D)D_t. \quad (8.2)$$

- ▶ There is a limited liability constraint, i.e., profits must be non-negative in all states:

$$\Psi_{\min}^I = (1 + i_t)B_t + (1 + i_t^X)B_t^g - (1 + i_t^D)D_t \geq 0. \quad (8.3)$$

- ▶ Banks maximize expected rents:

$$\mathcal{R}_t = E_t \left[\tilde{R}_{t,t+1} \Psi_{t+1}^I \right] - N_t, \quad (8.4)$$

where $\tilde{R}_{t,t+1}$ is the stochastic nominal discount factor of households.

Rents

- Substitute profits (8.2) into expression (8.4):

$$\mathcal{R}_t = E_t \left[\tilde{R}_{t,t+1} \left((1 + i_t)B_t + (1 + i_t^X)B_t^g - (1 + i_t^D)D_t \right) \right] - N_t.$$

- Intermediaries choose B_t, B_t^g, D_t, N_t , with $B_t^g, D_t, N_t \geq 0$ to maximize (8.4) under budget constraint (8.1), limited liability (8.3) and collateral constraint $B_t^g \geq \rho D_t$.
- We will use here also the household's Euler condition that prices the market interest rate i_t (discussed in details later):

$$E_t \{ \tilde{R}_{t,t+1} (1 + i_t) \} = 1.$$

Obtaining the Rent Function

We start from (8.4)

$$\mathcal{R}_t = E_t \left[\tilde{R}_{t,t+1} \Psi_{t+1}^I \right] - N_t.$$

We use equation (8.1) inside profits Ψ_{t+1}^I to obtain:

$$\Psi_{t+1}^I = (1 + i_t)(D_t + N_t - B_t^g) + (1 + i_t^X)B_t^g - (1 + i_t^D)D_t.$$

Rearranging:

$$\Psi_{t+1}^I = (1 + i_t)N_t + [(1 + i_t^X) - (1 + i_t)]B_t^g + [(1 + i_t) - (1 + i_t^D)]D_t.$$

Multiplying by $\tilde{R}_{t,t+1}$, taking expectations and using $E_t\{\tilde{R}_{t,t+1}(1 + i_t)\} = 1$ yields:

$$\mathcal{R}_t = \left[\frac{1 + i_t^X}{1 + i_t} - 1 \right] B_t^g - \left[\frac{1 + i_t^D}{1 + i_t} - 1 \right] D_t. \quad (8.5)$$

Optimization Problem

- ▶ Rent function (8.5) should be maximized under limited-liability constraint (8.3) and collateral constraint.
- ▶ Equity N_t can always be raised to satisfy limited liability $\Rightarrow$ constraint not binding in optimum.
- ▶ There are different cases for supply of deposits:
 - ▶ Case 1: Excess Reserves $B_t^g > \rho D_t$
 - ▶ Case 2: Binding collateral constraint $B_t^g = \rho D_t$ with special cases:
 - ▶ $\rho = 1$
 - ▶ $\rho = 0$

Case 1: Excess Reserves $B_t^g > \rho D_t$

- ▶ If collateral constraint is not binding, then B_t^g and D_t can be adjusted freely.
- ▶ If $\frac{1 + i_t^X}{1 + i_t} - 1 \neq 0$:
 - ▶ Rents $\mathcal{R}_t$ can be made arbitrarily large by adjusting B_t^g .
 - ▶ No equilibrium with finite rents.
- ▶ Therefore in an equilibrium with finite holdings:

$$\frac{1 + i_t^X}{1 + i_t} - 1 = 0 \quad \Rightarrow \quad i_t^X = i_t.$$

- ▶ Given $i_t^X = i_t$, zero rents also require

$$\frac{1 + i_t^D}{1 + i_t} - 1 = 0 \quad \Rightarrow \quad i_t^D = i_t^X = i_t.$$

- ▶ **Conclusion:** with excess reserves, all risk-free rates coincide.

Case 2: Binding collateral ($B_t^g = \rho D_t$)

- Substitute $B_t^g = \rho D_t$ into rent function (8.5):

$$\mathcal{R}_t = \left[\frac{1 + i_t^X}{1 + i_t} - 1 \right] \rho D_t - \left[\frac{1 + i_t^D}{1 + i_t} - 1 \right] D_t.$$

- Factor D_t :

$$\mathcal{R}_t = D_t \left\{ \rho \left(\frac{1 + i_t^X}{1 + i_t} - 1 \right) - \left(\frac{1 + i_t^D}{1 + i_t} - 1 \right) \right\}.$$

Deposit Rate

- ▶ Zero-rent condition ($\mathcal{R}_t = 0$ and $D_t > 0$) implies:

$$\rho \left(\frac{1 + i_t^X}{1 + i_t} - 1 \right) = \left(\frac{1 + i_t^D}{1 + i_t} - 1 \right).$$

- ▶ Rearranging:

$$1 + i_t^D = \rho(1 + i_t^X) + (1 - \rho)(1 + i_t). \quad (8.6)$$

- ▶ When rents are zero, the deposit rate is a weighted average of policy rate and credit-market rate.

Special Cases: $\rho = 1$ and $\rho = 0$

- ▶ **Narrow banking** ($\rho = 1$):

- ▶ Using (8.6) and $\rho = 1$ it follows that

$$1 + i_t^D = 1 + i_t^X \quad \Rightarrow \quad i_t^D = i_t^X.$$

- ▶ Generally $i_t \geq i_t^D = i_t^X$.

- ▶ **No collateral requirement** ($\rho = 0$):

- ▶ Equation (8.6) and $\rho = 0$ yields:

$$1 + i_t^D = 1 + i_t \quad \Rightarrow \quad i_t^D = i_t.$$

- ▶ When reserves are supplied at a positive level it must be that $i_t = i_t^X$.
- ▶ Therefore all risk-free rates coincide:

$$i_t = i_t^D = i_t^X.$$

Benchmark New-Keynesian Model as a special case

- ▶ The standard New-Keynesian model with a **single** nominal interest rate (policy rate = deposit rate = credit-market rate) is nested in this framework when either of the following holds:

- ① **Excess government liquidity:** the collateral constraint is slack,

$$B_t^g > \rho D_t,$$

so banks hold reserves in excess of what is required.

- ② **No collateral requirement / no liquidity services from B_t^g :**

$$\rho = 0.$$

- ▶ In both cases, all risk-free rates are equalized:

$$i_t^X = i_t^D = i_t.$$

Equity Demand

- ▶ Substituting the balance sheet (8.1) into the limited-liability constraint (8.3):

$$(1 + i_t)(D_t + N_t - B_t^g) + (1 + i_t^X)B_t^g - (1 + i_t^D)D_t \geq 0.$$

- ▶ Rearranging for N_t :

$$N_t \geq \frac{i_t^D - i_t}{1 + i_t} D_t + \frac{i_t - i_t^X}{1 + i_t} B_t^g.$$

- ▶ Using the expressions for the deposit rate and the collateral constraint, this implies

$$N_t \geq 0.$$

- ▶ Since all assets are riskless, equity is non-negative and can be zero in equilibrium.

Household Preferences

- Representative household maximizes:

$$E_{t_0} \left\{ \sum_{t=t_0}^{\infty} \beta^{t-t_0} \xi_t \left[U(C_t) + V\left(\frac{D_t}{P_t}\right) - \int_0^1 \frac{L_t(j)^{1+\eta}}{1+\eta} dj \right] \right\}. \quad (8.7)$$

- $U(\cdot)$: concave, differentiable utility from consumption C_t (Dixit-Stiglitz aggregator).
 - $V(\cdot)$: concave, differentiable utility from real deposits D_t/P_t .
 - $V_d(\cdot) = 0$ for $D_t/P_t \geq \bar{d}$ (satiation level $\bar{d}$).
 - ξ_t : preference shock.
 - $\eta \geq 0$: inverse of the Frisch elasticity of labor.
- The only difference to benchmark NK model is that households get utility from deposits, through liquidity term $V(\cdot)$.

Household Budget Constraint

- ▶ Flow budget constraint:

$$P_t C_t + D_t + (1 + i_{t-1})B_{t-1} + N_t + T_t \leq (1 + i_{t-1}^D)D_{t-1} + B_t + \int_0^1 W_t(j)L_t(j)dj + \Psi_t^I + \Psi_t.$$

- ▶ LHS: consumption, new deposits, repayment of bonds, new equity, taxes.
- ▶ RHS: deposit returns, new bonds, labor income, profits from intermediaries (Ψ_t^I) and firms (Ψ_t).
- ▶ Households trade:
 - ▶ Illiquid, risk-free bonds B_t at rate i_t ($B_t > 0$ denotes debt),
 - ▶ Deposits D_t at rate i_t^D , providing liquidity services,
 - ▶ Equity N_t for banks.
- ▶ The consumption/saving choices are subject to an appropriate borrowing limit.

Euler Equation for Illiquid Bonds

- Optimal choice of B_t implies the condition:

$$E_t\{\tilde{R}_{t,t+1}\} = \frac{1}{1 + i_t}, \quad (8.8)$$

where the nominal stochastic discount factor is:

$$\tilde{R}_{t,t+1} = \beta \frac{\xi_{t+1} U_c(C_{t+1})}{\xi_t U_c(C_t)} \frac{P_t}{P_{t+1}}.$$

- This is the standard NK asset-pricing condition:

$$1 = E_t [\tilde{R}_{t,t+1}(1 + i_t)].$$

- i_t is the *credit-market nominal interest rate* relevant for consumption-savings.

FOC for Deposits

- FOC for deposits D_t :

$$1 = \varrho_t + (1 + i_t^D)E_t\{\tilde{R}_{t,t+1}\}, \quad (8.9)$$

where ϱ_t is the liquidity premium:

$$\varrho_t = \frac{V_d(\frac{D_t}{P_t})}{U_c(C_t)}.$$

- Combining (8.8) with (8.9) we obtain:

$$1 + i_t^D = (1 - \varrho_t)(1 + i_t). \quad (8.10)$$

Liquidity Satiation and Equity

- ▶ With $0 \leq \varrho_t < 1$:

- ▶ Deposits pay a lower interest rate than illiquid bonds:

$$i_t^D \leq i_t.$$

- ▶ The wedge $i_t - i_t^D$ is determined by liquidity services.
 - ▶ If the economy is satiated with liquidity, i.e., $\varrho_t = 0$ then the two rates coincide ($i_t^D = i_t$)
 - ▶ FOC for equity:

$$N_t = E_t \{ \tilde{R}_{t,t+1} \Psi_{t+1}^I \}.$$

- ▶ Finally, the intertemporal budget constraint of the consumer holds with equality at all times.

Firms: Technology and Pricing

- ▶ Continuum of firms $j \in [0, 1]$ produce $Y_t(j) = A_t L_t(j)$, with labor productivity A_t .
- ▶ Demand for variety j is $Y_t(j) = \left(\frac{p_t(j)}{P_t}\right)^{-\theta} Y_t$, with elasticity of substitution $\theta > 1$.
- ▶ The aggregate price index P_t is the Dixit–Stiglitz aggregator.
- ▶ Calvo price setting:
 - ▶ Fraction $1 - \alpha$ can reset prices each period.
 - ▶ Non-adjusting firms index prices to target inflation Π .

AS Block

- As in the benchmark NK model of Chapter 5 (with the only difference that here $G_t = 0$), equilibrium pricing conditions yield:

$$\left(\frac{1 - \alpha(\Pi_t/\Pi)^{\theta-1}}{1 - \alpha} \right)^{\frac{1+\theta\eta}{\theta-1}} = \frac{F_t}{J_t},$$

with

$$F_t = \xi_t U_c(Y_t) \frac{Y_t}{\mu_t} + \alpha\beta E_t \left\{ \left(\frac{\Pi_{t+1}}{\Pi} \right)^{\theta-1} F_{t+1} \right\},$$
$$J_t = \xi_t \left(\frac{Y_t}{A_t} \right)^{1+\eta} + \alpha\beta E_t \left\{ \left(\frac{\Pi_{t+1}}{\Pi} \right)^{\theta(1+\eta)} J_{t+1} \right\}.$$

Government

- ▶ Consolidate treasury and central bank.
- ▶ Flow budget constraint:

$$B_t^g = (1 + i_{t-1}^X)B_{t-1}^g + \tau_t^s P_t Y_t - T_t, \quad (8.11)$$

where

- ▶ B_t^g : total government liabilities (including reserves),
 - ▶ τ_t^s : subsidy to firm revenues (markup correction),
 - ▶ T_t : lump-sum taxes.
-
- ▶ Central bank fully backs treasury, therefore overall government debt is not subject to a solvency condition.

- 1 Introduction
- 2 Model
- 3 Equilibrium**
- 4 Log-linear approximation
- 5 Impulse Responses
- 6 Optimal policy
- 7 Main Takeaways
- 8 Bibliography

Equilibria in Assets and Goods Markets

- ▶ Asset-market equilibrium:
 - ▶ All government bonds B^g held by intermediaries.
 - ▶ Deposits D are held by the households.
 - ▶ B issued by the households is an asset for the intermediaries.
- ▶ Goods market equilibrium implies $Y_t = C_t$ for each $t \geq t_0$.

AD Block

- Recall the relationship between the deposit-, policy- and credit-market rate (8.6) (binding collateral):

$$1 + i_t^D = \rho(1 + i_t^X) + (1 - \rho)(1 + i_t). \quad (8.12)$$

- From (8.10), the spread between the deposit rate and credit-market rate must satisfy

$$\frac{1 + i_t^D}{1 + i_t} = 1 - \frac{V_d(D_t/P_t)}{U_c(Y_t)}, \quad (8.13)$$

where, using the definition of the stochastic discount factor, (8.8) implies:

$$\frac{1}{1 + i_t} = E_t \left\{ \beta \frac{\xi_{t+1} U_c(Y_{t+1})}{\xi_t U_c(Y_t)} \frac{P_t}{P_{t+1}} \right\}. \quad (8.14)$$

AS Block

- As already stated, the AS block of the model is as in Chapter 5:

$$\left(\frac{1 - \alpha(\Pi_t/\Pi)^{\theta-1}}{1 - \alpha} \right)^{\frac{1+\theta\eta}{\theta-1}} = \frac{F_t}{J_t}, \quad (8.15)$$

with

$$F_t = \xi_t U_c(Y_t) \frac{Y_t}{\mu_t} + \alpha\beta E_t \left\{ \left(\frac{\Pi_{t+1}}{\Pi} \right)^{\theta-1} F_{t+1} \right\}, \quad (8.16)$$

$$J_t = \xi_t \left(\frac{Y_t}{A_t} \right)^{1+\eta} + \alpha\beta E_t \left\{ \left(\frac{\Pi_{t+1}}{\Pi} \right)^{\theta(1+\eta)} J_{t+1} \right\}, \quad (8.17)$$

and markup $\mu_t = \frac{\theta}{(\theta-1)(1+\tau_t^s)}$ where τ_t^s is the tax/subsidy.

Constraints

- ▶ Government flow budget constraint (8.12):

$$B_t^g = (1 + i_{t-1}^X)B_{t-1}^g + \tau_t^s P_t Y_t - T_t. \quad (8.18)$$

- ▶ Intertemporal resource constraint:

$$\frac{(1 + i_{t-1}^X)B_{t-1}^g}{P_t} = E_t \left\{ \sum_{T=t}^{\infty} \beta^{T-t} \frac{\xi_T U_c(Y_T)}{\xi_t U_c(Y_t)} \left[\frac{T_T}{P_T} - \tau_T^s Y_T + \frac{i_T - i_T^X}{1 + i_T} \frac{B_T^g}{P_T} \right] \right\}. \quad (8.19)$$

- ▶ Collateral constraint:

$$B_t^g \geq \rho D_t. \quad (8.20)$$

Equilibrium and Policy Regime

- An equilibrium is a **set of stochastic processes**

$$\{P_t, i_t, i_t^D, i_t^X, Y_t, F_t, J_t, D_t, T_t, B_t^g\}_{t=t_0}^\infty,$$

satisfying:

- Equations (8.12)–(8.18), (8.20), for each $t \geq t_0$ and (8.19) considering
 - Inflation definition $\Pi_t \equiv P_t/P_{t-1}$,
 - Inequality constraint $i_t \geq i_t^X \geq 0$,
 - Given exogenous processes $\{\xi_t, A_t, \tau_t^s\}_{t=t_0}^\infty$ and initial conditions.
-
- **Policy degrees of freedom:**
 - Central bank sets the interest rate on reserves $\{i_t^X\}_{t=t_0}^\infty$,
 - Central bank and treasury jointly set total government liabilities $\{B_t^g\}_{t=t_0}^\infty$ (including reserves).

Relation to the Benchmark NK Model

- ▶ Key departure from the benchmark NK model (Chapter 5):
 - ▶ **Government debt** directly influences aggregate demand,
 - ▶ Transmission occurs through equations (8.13) and (8.20).
- ▶ The standard NK model is **nested** as a special case:
 - ▶ When liquidity is fully satiated:

$$V_d(\cdot) = 0.$$

- 1 Introduction
- 2 Model
- 3 Equilibrium
- 4 Log-linear approximation**
- 5 Impulse Responses
- 6 Optimal policy
- 7 Main Takeaways
- 8 Bibliography

Log-linearization: Notation and Focus

- ▶ Log-deviation (unless otherwise noted): $\hat{X}_t \equiv \ln(X_t/X)$.
- ▶ Inflation: $\Pi_t \equiv P_t/P_{t-1}$, $\pi_t \equiv \ln \Pi_t$, target $\pi \equiv \ln \Pi$.
- ▶ Real deposits: $d_t \equiv D_t/P_t$, $\hat{d}_t \equiv \ln(d_t/d)$.
- ▶ Interest-rate deviations:

$$\hat{i}_t \equiv \ln\left(\frac{1+i_t}{1+i}\right), \quad \hat{i}_t^X \equiv \ln\left(\frac{1+i_t^X}{1+i^X}\right), \quad \hat{i}_t^D \equiv \ln\left(\frac{1+i_t^D}{1+i^D}\right).$$

- ▶ Goal: express the NK system in log-linear form, highlighting the role of **(i) multiple rates** and **(ii) liquidity (deposits)**.

New-Keynesian Phillips curve

- Log-linear AS equation (NKPC):

$$\pi_t - \pi = \kappa(\hat{Y}_t - \hat{Y}_t^n) + \beta E_t(\pi_{t+1} - \pi). \quad (8.21)$$

- Natural output:

$$\hat{Y}_t^n = \frac{1 + \eta}{\sigma^{-1} + \eta} \hat{A}_t - \frac{1}{\sigma^{-1} + \eta} \hat{\mu}_t, \quad \sigma \equiv -\frac{U_{cc}Y}{U_c} > 0.$$

- Benchmark NK AS equation holds.

AD in Terms of the Credit-Market Rate

- Log-linearizing the Euler equation (8.14):

$$\hat{Y}_t = E_t \hat{Y}_{t+1} - \sigma \left(\hat{i}_t - E_t(\pi_{t+1} - \pi) + E_t \Delta \hat{\xi}_{t+1} \right). \quad (8.22)$$

- Preference growth term: $\Delta \hat{\xi}_{t+1} \equiv \hat{\xi}_{t+1} - \hat{\xi}_t$.
- Key difference vs benchmark NK: (8.22) depends on the **credit-market rate** i_t (not directly the policy rate).

Money Market Rates

- ▶ Banking sector implies the following (log-linear) relationship:

$$\hat{i}_t = \hat{i}_t^X + \frac{1 - \nu}{\rho - \nu} (\hat{i}_t - \hat{i}_t^D), \quad (8.23)$$

where $\nu \equiv V_d/U_c$ (evaluated at steady state) and $\nu < \rho$.

- ▶ Interpretation:
 - ▶ The policy rate i_t^X transmits to the economy through bank pricing of deposits.
 - ▶ The spread $(\hat{i}_t - \hat{i}_t^D)$ matters for the mapping from policy to the credit rate.

Derivation of Money Market Rates and Deposit Demand (Appendix)

Liquidity Demand

- Liquidity market equilibrium implies:

$$\hat{d}_t = d_y \hat{Y}_t - d_i (\hat{i}_t - \hat{i}_t^D), \quad (8.24)$$

where

$$d_y \equiv \frac{\sigma_d}{\sigma}, \quad d_i \equiv \sigma_d \frac{1 - \nu}{\nu}, \quad \sigma_d \equiv -\frac{V_d}{V_{dd}d}.$$

- Higher output raises liquidity demand; a higher spread $(i_t - i_t^D)$ lowers liquidity demand.

Benchmark NK as a Special Case (Full Satiation)

- ▶ If liquidity is fully satiated: $\nu = 0$.
- ▶ Then rates equalize in and out of steady state:

$$i_t^D = i_t^X = i_t.$$

- ▶ AD collapses to the benchmark NK AD curve:

$$\hat{Y}_t = E_t \hat{Y}_{t+1} - \sigma \left(\hat{i}_t^X - E_t(\pi_{t+1} - \pi) + E_t \Delta \hat{\xi}_{t+1} \right). \quad (8.25)$$

Aggregate Demand with Liquidity

- In the general case ($\nu > 0$), combining (8.22), (8.23), (8.24) yields:

$$\hat{Y}_t = v_\rho E_t \hat{Y}_{t+1} - \sigma v_\rho \left(\hat{i}_t^X - E_t(\pi_{t+1} - \pi) + E_t \Delta \hat{\xi}_{t+1} \right) + d_y^{-1} (1 - v_\rho) \hat{d}_t, \quad (8.26)$$

where

$$v_\rho \equiv 1 - \rho^{-1} \nu \in (0, 1).$$

Main Implications vs the Benchmark NK Model

- ▶ **Liquidity shifts AD directly:** deposits $\hat{d}_t$ enter (8.26).
- ▶ **Attenuated forward guidance:** $\nu_\rho < 1$ reduces the effect of expected future output (and thus expected future policy) on current output.
- ▶ **Two transmission margins:**
 - ▶ Policy rate i_t^X affects demand through the banking-sector wedge (8.23),
 - ▶ Liquidity d_t affects demand directly (8.26).
- ▶ Higher deposits —corresponding to a larger supply of government liquidity— are **expansionary** for output.

- 1 Introduction
- 2 Model
- 3 Equilibrium
- 4 Log-linear approximation
- 5 Impulse Responses**
- 6 Optimal policy
- 7 Main Takeaways
- 8 Bibliography

Impulse Response to a Policy-Rate Shock

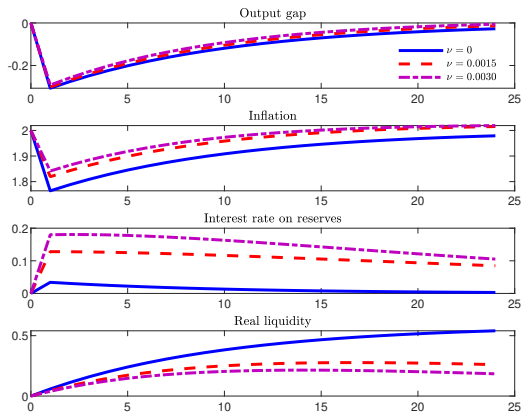


Figure 1: Impulse Response to a Shock to the Interest Rate on Reserves

- Monetary policy follows the interest-rate rule
$$\hat{i}_t^X = \phi_\pi (\pi_t - \pi) + \phi_y \hat{Y}_t + e_t$$
with $e_t = \rho_e e_{t-1} + \varepsilon_{1,t}$ and $0 \leq \rho_e \leq 1$ as well as $\phi_\pi, \phi_y \geq 0$.
- The Figure shows the responses of output gap, inflation, interest rate on reserves, real liquidity to a one-time shock $\varepsilon_{1,t}$.

Impulse Response to a Policy Shock

- ▶ The figure shows the impulse responses for three different calibrations of $\nu \in \{0, 0.0015, 0.003\}$ which implies interest rate spreads of $(i - i^D) \in \{0, 60, 120\}$ in basis points at annual rates.
- ▶ When $\nu = 0$, the benchmark NK model is nested:
 - ▶ A positive interest-rate shock is contractionary, lowering output gap and inflation.
 - ▶ Liquidity does not affect aggregate demand; real liquidity rises only mechanically as inflation falls.
- ▶ When $\nu > 0$, liquidity influences aggregate demand:
 - ▶ Higher real liquidity mitigates the decline in the output gap and inflation.
 - ▶ The stabilizing effect is present but quantitatively small.
- ▶ Because inflation falls less when $\nu > 0$, real liquidity increases by less than in the $\nu = 0$ case.

Impulse Response to a Liquidity Shock

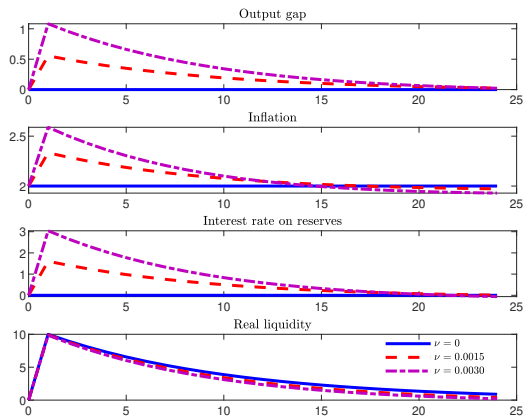


Figure 2: Impulse Response to a Shock in the Overall Liquidity Supply

- ▶ Nominal government debt in deviations from the steady state follows the process $\hat{B}_t^g = \rho_b \hat{B}_{t-1}^g + \varepsilon_{2,t}$ with $0 \leq \rho_b \leq 1$.
- ▶ In equilibrium $\hat{B}_t^g = \rho \hat{d}_t$.
- ▶ The Figure shows the responses of output gap, inflation, interest rate on reserves, real liquidity to a one-time shock $\varepsilon_{2,t}$.

Impulse Response to a Liquidity Shock

- ▶ With $\nu > 0$, liquidity directly affects aggregate demand:
 - ▶ Higher liquidity raises the output gap through the AD equation,
 - ▶ This translates into higher inflation.
- ▶ Monetary and fiscal policies affect the economy not only through interest rates:
 - ▶ Government debt and central bank reserves influence economic activity,
 - ▶ They complement conventional interest-rate policy.
- ▶ Expanding government liquidity boosts demand:
 - ▶ Higher government liquidity raises deposits, lowers the liquidity premia and the credit-market interest rate, and pushes consumption and output up,
 - ▶ Increased economic activity feeds into inflation via the AS equation (8.21).

- 1 Introduction
- 2 Model
- 3 Equilibrium
- 4 Log-linear approximation
- 5 Impulse Responses
- 6 Optimal policy**
- 7 Main Takeaways
- 8 Bibliography

Optimal Liquidity Policy

- ▶ Optimal policy maximizes household welfare, which now includes utility from holding real deposits.
- ▶ With lump-sum taxes, optimal policy **satiates liquidity**:

$$V_d(\cdot) = 0,$$

eliminating liquidity premia in equilibrium.

- ▶ Friedman (1960) logic:
 - ▶ The optimal allocation closes the wedge between money-market rates and the policy rate.
 - ▶ When liquidity is interest-bearing, full satiation does **not** restrict the choice of the inflation target (unlike in Chapter 2).
- ▶ With distortionary taxes, full liquidity satiation is no longer optimal:
 - ▶ A second-best trade-off arises between liquidity provision and tax distortions.

Optimal Policy with Lump-Sum Taxes

- ▶ We focus on optimal stabilization when lump-sum taxes are available and the steady-state subsidy τ_t^s offsets monopolistic distortions.
- ▶ Liquidity assumptions near satiation:
 - ▶ The economy is close to first-best liquidity satiation (V_d small, v small).
 - ▶ Liquidity demand remains well defined below the satiation level.
 - ▶ In the limit, deposit demand retains the form of (8.24) with $d_y = 0$ and $d_i = -U_c/(V_{dd}d)$.

Quadratic Loss Function with Liquidity

- Under the assumptions stated above, the second-order approximation of household utility implies the loss function:

$$E_{t_0} \left\{ \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left[\frac{1}{2} (\hat{Y}_t - \hat{Y}_t^e)^2 + \frac{1}{2} \Lambda_d (\hat{d}_t - d^*)^2 + \frac{1}{2} \frac{\theta}{\kappa} (\pi_t - \pi)^2 \right] \right\}. \quad (8.27)$$

- The policymaker trades off stabilization of the output gap $(\hat{Y}_t - \hat{Y}_t^e)$, inflation deviations $(\pi_t - \pi)$ and real liquidity deviations $(\hat{d}_t - d^*)$.
- The liquidity target $d^* = \nu d_i$ captures steady-state liquidity distortions and is of the same order as the liquidity parameter ν .
- In the first best case, liquidity is fully satiated and $\hat{d}_t = d^*$.

Derivation of Quadratic Loss Function with Liquidity (Appendix)

Optimal Stabilization Policy

- ▶ The policymaker minimizes the quadratic loss function subject to the **AS equation**:

$$\pi_t - \pi = \kappa(\hat{Y}_t - \hat{Y}_t^e) + v_t + \beta E_t(\pi_{t+1} - \pi), \quad (8.28)$$

where

$$v_t = \frac{\kappa}{\sigma^{-1} + \eta} \hat{\mu}_t, \quad \hat{Y}_t^e = \frac{1 + \eta}{\sigma^{-1} + \eta} \hat{A}_t.$$

- ▶ The minimization problem is also subject to the **AD equation**. When ν is of a small order:
 - ▶ Equilibrium in the liquidity market remains of the same form as in (8.24), with $d_y = 0$ and $d_i = -\frac{U_c}{V_{dd}d}$.
 - ▶ The functional form of the AD equation (8.22) is unchanged while $\nu = 0$ in (8.23).

Optimal Liquidity and Stabilization Policy

- In the limit $\nu \rightarrow 0$, the aggregate demand equation (8.26) becomes:

$$\hat{Y}_t = E_t \hat{Y}_{t+1} - \sigma \left(\hat{i}_t^X - E_t(\pi_{t+1} - \pi) + E_t \Delta \hat{\xi}_{t+1} \right) + \rho^{-1} d_i^{-1} \sigma \hat{d}_t. \quad (8.29)$$

- **Optimal liquidity policy:** liquidity is fully satiated, so that

$$d_t = d^* \quad \text{at all times.}$$

- With liquidity optimally set, the remaining policy problem coincides with the standard NK stabilization problem.

Optimal Policy and Liquidity

- ▶ Combining the targeting rule with the AS equation (8.28) yields the **same optimal response to shocks** as in Chapter 7.
- ▶ The key reason is that liquidity is an **additional policy instrument**: under optimal policy it is always set to its efficient level,

$$d_t = d^*,$$

closing liquidity premia.

- ▶ As a result:
 - ▶ The steady-state spread between money-market rates and the policy rate is eliminated.
 - ▶ All risk-free rates move together, as implied by the AD equation (8.29).
- ▶ In the absence of markup shocks, the **first best** is attainable: output, inflation, and liquidity deviations can be simultaneously stabilized in the loss function (8.27).

Broader Implications

- ▶ In the presence of markup shocks, a trade-off between output and inflation stabilization emerges, but it remains optimal to **fully satiate liquidity** and eliminate money-market spreads.
- ▶ Liquidity frictions therefore matter primarily when policy is **sub-optimal**, as illustrated by the responses in Figure 2.
- ▶ From a normative perspective, the policymaker uses government liquidity to operate **as close as possible to the benchmark NK allocation**, neutralizing financial frictions whenever feasible.
- ▶ Chapter 9 introduces an additional friction—**distortionary taxation**— which limits optimal liquidity provision and fundamentally alters the nature of stabilization policy.

- 1 Introduction
- 2 Model
- 3 Equilibrium
- 4 Log-linear approximation
- 5 Impulse Responses
- 6 Optimal policy
- 7 Main Takeaways**
- 8 Bibliography

Main Takeaways

- ▶ **Banking and interest rates:** introducing a banking sector generates a **hierarchy of risk-free rates**; liquidity premia drive wedges between the policy rate, deposit rate, and credit-market rate.
- ▶ **Government liquidity:** reserves and treasury bills affect **aggregate demand** by influencing liquidity premia, acting as a transmission channel beyond the standard interest-rate channel.
- ▶ **Liquidity satiation:** when liquidity is fully satiated, the model collapses to the **benchmark New Keynesian framework** and monetary policy operates solely through the policy rate.
- ▶ **Optimal policy:** under lump-sum taxes, liquidity is optimally satiated at all times; the **optimal stabilization policy** coincides with standard results of the benchmark NK framework.
- ▶ **Frictions and distortions:** liquidity frictions matter primarily under sub-optimal policies or **distortionary taxation**, which constrain the government's ability to provide optimal liquidity.

- 1 Introduction
- 2 Model
- 3 Equilibrium
- 4 Log-linear approximation
- 5 Impulse Responses
- 6 Optimal policy
- 7 Main Takeaways
- 8 Bibliography**

References



Friedman, M. (1960). *A program for monetary stability*. Fordham University Press.

9 Appendix

Appendix: steady state

- Deterministic steady state with constant shocks:

$$\xi_t = \xi, \quad A_t = A, \quad \mu_t = 1.$$

- Markup distortion eliminated by subsidy:

$$\tau^s = \frac{1}{\theta - 1}.$$

- Inflation at target: $\Pi_t = \Pi$.
- AS block (8.15)–(8.17) implies:

$$F = J, \quad F = \frac{\xi U_c(Y)Y}{1 - \alpha\beta}, \quad J = \frac{\xi(Y/A)^{1+\eta}}{1 - \alpha\beta}.$$

- Efficient steady-state output:

$$\left(\frac{Y}{A}\right)^\eta = AY U_c(Y).$$

Appendix: steady state

- ▶ Euler equation (8.14) with $\Pi_t = \Pi$:

$$1 + i = \beta^{-1}\Pi.$$

- ▶ Assume $D/P < \bar{d}$ and collateral binds: $B^g = \rho D$.
- ▶ Define $v \equiv V_d/U_c$ in steady state. Then:

$$1 + i^D = (1 - v)(1 + i). \quad (??)$$

$$1 + i^D = \rho(1 + i^X) + (1 - \rho)(1 + i). \quad (??)$$

- ▶ Combining:

$$\frac{1 + i}{1 + i^X} = \frac{\rho}{\rho - v}, \quad \frac{1 + i^D}{1 + i^X} = \frac{\rho(1 - v)}{\rho - v}, \quad (v < \rho) \Rightarrow i > i^D > i^X.$$

Appendix: derivation of the log-linear approximation

- Start from (8.12) and subtract steady state:

$$i_t^D - i^D = \rho(i_t^X - i^X) + (1 - \rho)(i_t - i).$$

- Divide by $(1 + i^D)$ and use first-order log deviations:

$$\hat{i}_t^D = \rho \frac{1 + i^X}{1 + i^D} \hat{i}_t^X + (1 - \rho) \frac{1 + i}{1 + i^D} \hat{i}_t. \quad (??)$$

- Using steady-state ratios (via ν) implies:

$$\hat{i}_t^D = \frac{\rho - \nu}{1 - \nu} \hat{i}_t^X + \frac{1 - \rho}{1 - \nu} \hat{i}_t \Rightarrow \hat{i}_t = \hat{i}_t^X + \frac{1 - \nu}{\rho - \nu} (\hat{i}_t - \hat{i}_t^D). \quad (8.23)$$

Appendix: deposit demand

- First-order approximation of (8.13) implies:

$$(1 - \nu)(\hat{i}_t^D - \hat{i}_t) = -\frac{V_{dd}d}{V_d}\nu\hat{d}_t - \frac{U_{cc}Y}{U_c}\nu\hat{Y}_t.$$

- Using $\sigma_d \equiv -\frac{V_d}{V_{dd}d}$ and $\sigma \equiv -\frac{U_{cc}Y}{U_c}$:

$$\hat{d}_t = d_y\hat{Y}_t - d_i(\hat{i}_t - \hat{i}_t^D), \quad d_y = \frac{\sigma_d}{\sigma}, \quad d_i = \sigma_d \frac{1 - \nu}{\nu}. \quad (8.24)$$

Appendix: AD equation, forward solution

- Solving (8.26) forward:

$$\begin{aligned}\hat{Y}_t = & -v_\rho \sigma E_t \sum_{T=t}^{\infty} v_\rho^{T-t} \left(\hat{l}_T^X - (\pi_{T+1} - \pi) + \Delta \hat{\xi}_{T+1} \right) \\ & + d_y^{-1} (1 - v_\rho) E_t \sum_{T=t}^{\infty} v_\rho^{T-t} \hat{d}_T.\end{aligned}\tag{8.30}$$

- $v_\rho < 1$ implies geometrically decaying weights on future policy and future liquidity.

[Back to Log-Linear Rate Mapping](#)

Appendix: Loss Function Derivation

- ▶ The loss function (8.27) is obtained as a second-order approximation of household utility (8.7), following Chapter 7.
- ▶ We focus on the liquidity term $V(d_t)$, assuming:
 - ▶ The economy is close to liquidity satiation: $V_d = O(\|\xi\|)$,
 - ▶ The liquidity parameter ν is of small order.
- ▶ As d_t approaches the satiation level $\bar{d}$ from below:
 - ▶ $V_{dd} < 0$, ensuring a well-defined liquidity demand.
 - ▶ In the limit $V_d \rightarrow 0$, liquidity demand retains the form of (8.24) with

$$d_y = 0, \quad d_i = -\frac{U_c}{V_{dd}\bar{d}}.$$

Appendix: Loss Function Derivation

Consider a second-order Taylor expansion of $V(d_t)$:

$$V(d_t) = V(d) + V_d(d_t - d) + \frac{1}{2}V_{dd}(d_t - d)^2 + \mathcal{O}(\|\xi\|^3).$$

Using

$$(d_t - d) = d\left(\hat{d}_t + \frac{1}{2}\hat{d}_t^2\right) + \mathcal{O}(\|\xi\|^3),$$

we obtain:

$$V(d_t) = V(d) + V_d d \hat{d}_t + \frac{1}{2}(V_d d + V_{dd} d^2) \hat{d}_t^2 + \mathcal{O}(\|\xi\|^3).$$

Neglecting terms independent of policy:

$$V(d_t) = U_c Y \left[\frac{V_d d}{U_c Y} \hat{d}_t + \frac{1}{2} \frac{V_{dd} d^2}{U_c Y} \hat{d}_t^2 \right] + \mathcal{O}(\|\xi\|^3).$$

Appendix: Loss Function Derivation

Using

$$v \equiv \frac{V_d}{U_c}, \quad d_i \equiv -\frac{U_c}{V_{dd}d},$$

we can write:

$$V(d_t) = U_c Y \frac{d}{Y} \left[v \hat{d}_t - \frac{1}{2} \frac{1}{d_i} \hat{d}_t^2 \right] + \mathcal{O}(\|\xi\|^3).$$

Conclude that:

$$V(d_t) = -\frac{1}{2} \frac{U_c d}{d_i} (\hat{d}_t - d^*)^2 + \mathcal{O}(\|\xi\|^3),$$

where $d^* \equiv v d_i$.

Appendix: Loss Function Derivation

- Result from chapter 7 (utility approximation):

$$U_{t_0} = -(\sigma^{-1} + \eta)L^{1+\eta}E_{t_0} \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left[\frac{1}{2}(\hat{Y}_t - \hat{Y}_t^e)^2 + \frac{\theta}{2\kappa}(\pi_t - \pi)^2 \right] + \text{t.i.p.}$$

- Combining this expression with the previously derived liquidity term and defining $\Delta \equiv \frac{d}{Yd_i}$ yields the quadratic loss function (8.27):

$$E_{t_0} \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left[\frac{1}{2}(\hat{Y}_t - \hat{Y}_t^e)^2 + \frac{1}{2}\Delta(\hat{d}_t - d^*)^2 + \frac{\theta}{2\kappa}(\pi_t - \pi)^2 \right],$$

which corresponds to equation (8.27).

[Back to Loss Function](#)